Mechanical & Industrial Engineering
UNIVERSITY OF TORONTO

CPAIOR 2018

Modelling and Solving the Senior Transportation Problem

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CHATS

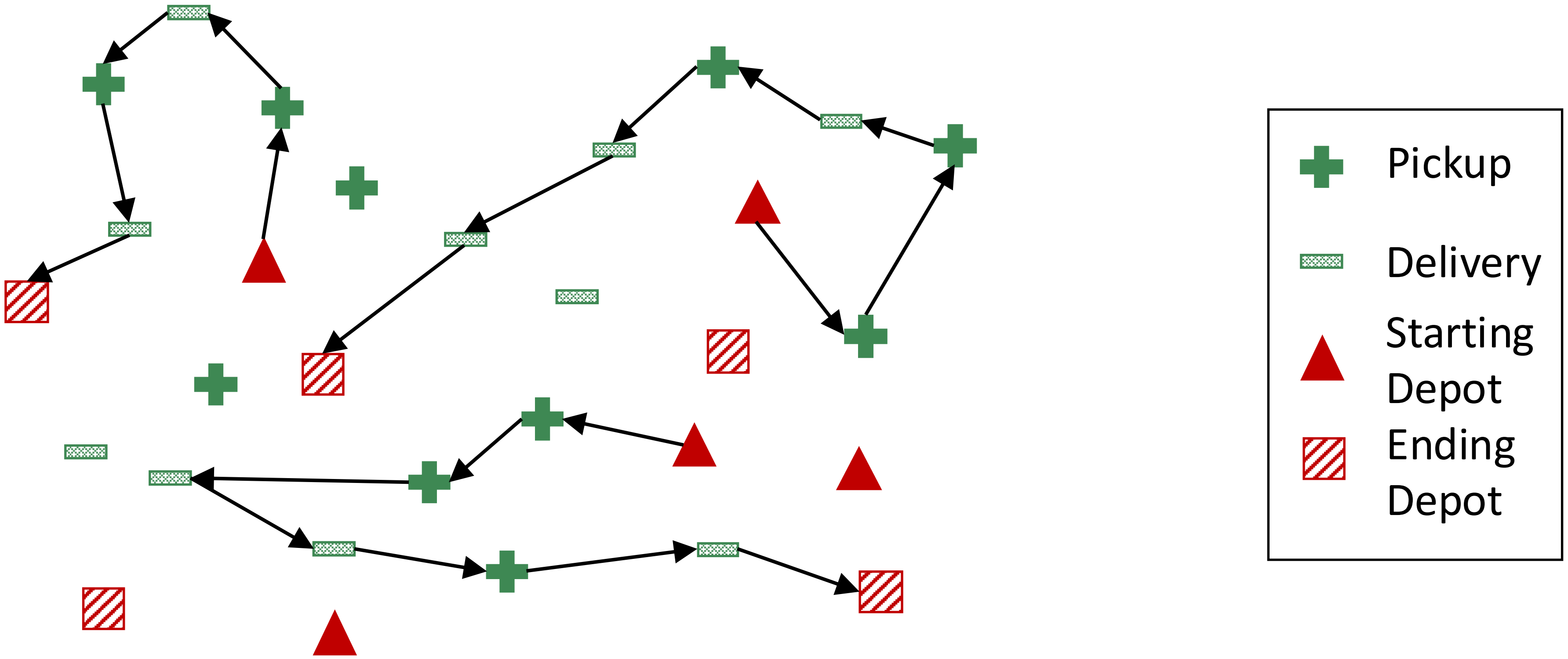
(Community & Home Assistance to Seniors)

- Provided **88,000** trips to **7300+** seniors in 2015
- **4%** of daily trips not scheduled resulting in **3500** unmet requests yearly
- Vehicles are not used under full capacity **85%** of the time

3,346 km² 315 km

The Senior Transportation Problem

3. Schedule each route



Solution Approaches

Exact Techniques

MIP (mixed integer programming)	CP (constraint programming)
MIP/CP LBBD (logic-based Benders decomposition)	CP/CP LBBD (logic-based Benders decomposition)

Non-exact Technique

construction heuristic

Variables for the MIP model

$$x_{k,i,j} = \begin{cases} 1, & \text{if vehicle k visits location j \underline{directly} after visiting location i} \\ 0, & \text{otherwise} \end{cases}$$

$$v_{k,i} = \text{load of vehicle k after visiting location i}$$

$$u_{k,i} = \text{time when vehicle k leaves location i after completing the service at location i}$$

MIP Formulation

$$\max \sum \sum \sum \left(W_r \times x_{k,i_{r+},j}\right) \quad (1)$$

$$\max \quad \text{s.t.} \quad \sum_{j \in \mathcal{N}^+} x_{k,i_{k+},j} + x_{k,i_{k+},i_{k-}} = 1 \qquad \forall k \in \mathcal{K} \qquad (2)$$

$$\sum_{i \in \mathcal{N}^-} x_{k,i,j_{k-}} + x_{k,i_{k+},i_{k-}} = 1 \qquad \forall k \in \mathcal{K} \qquad (3)$$

$$u_{k,i} \geq E_i - M \times \left(1 - \sum_{j \in \mathcal{V}} x_{k,i,j}\right) \qquad \forall k \in \mathcal{K}, i \in \mathcal{V} \qquad (8)$$

$$u_{k,i} \leq L_i - S_i + M \times \left(1 - \sum_{j \in \mathcal{V}} x_{k,i,j}\right) \qquad \forall k \in \mathcal{K}, i \in \mathcal{V} \qquad (9)$$

$$v_{k,j} \geq (v_{k,i} + Q_i) - M \times (1 - x_{k,i,j}) \qquad \forall k \in \mathcal{K}, i, j \in \mathcal{V} \qquad (12)$$

All trips must stay within the specified time windows and limited capacity

$$u_{k,i} \geq E_i - M \times \left(1 - \sum_{j \in \mathcal{V}} x_{k,i,j}\right) \qquad \forall k \in \mathcal{K}, i \in \mathcal{V} \qquad (8)$$

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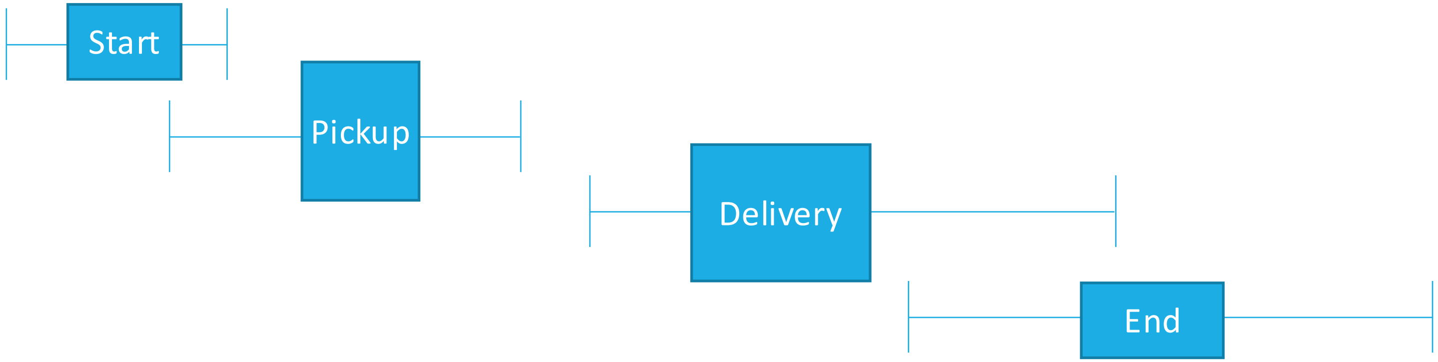
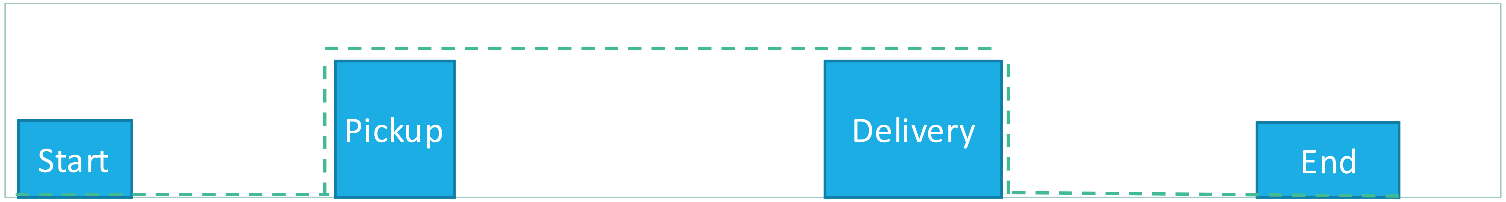
Interval Variables for the CP Model

Each location variable is represented by an **interval variable**



Each route is a **sequence variable**



CP Formulation

$$\text{maximize} \sum_{r \in \mathcal{R}} (W_r \times \texttt{IloPresenceOf}(x_i))$$

subject to

<u>Location Constraints</u>

$$\texttt{IloAlternative}(x_i, X_i) \qquad \forall i \in \mathcal{N}$$

$$\texttt{IloBefore}(\bar{X}_{k,i_{r+}}, \bar{X}_{k,i_{r-}}) \qquad \forall k \in \mathcal{K}, \forall r \in \mathcal{R}$$

$$\texttt{IloPresenceOf}(\bar{X}_{k,i_{r+}}) = \texttt{IloPresenceOf}(\bar{X}_{k,i_{r-}}) \qquad \forall k \in \mathcal{K}, \forall r \in \mathcal{R}$$

<u>Ride Time Constraint</u>

$$\texttt{GetStartMax}(x_{i_{r-}}) - \texttt{GetEndMax}(x_{i_{r+}}) \leq Z \qquad \forall r \in \mathcal{R}$$

<u>Capacity Constraints</u>

$$y_{k,i} = \texttt{IloStepAtStart}(\bar{X}_{k,i}, Q_i) \qquad \forall k \in \mathcal{K}, \forall i \in \mathcal{N}$$

$$\sum_{i \in \mathcal{N}} y_{k,i} \leq C_k \qquad \forall k \in \mathcal{K}$$

<u>Route Constraints</u>

$$\texttt{IloFirst}(u_k, x_{i_{k+}}) \qquad \forall k \in \mathcal{K}$$

$$\texttt{IloLast}(u_k, x_{i_{k-}}) \qquad \forall k \in \mathcal{K}$$

$$\texttt{IloNoOverlap}(u_k) \qquad \forall k \in \mathcal{K}$$

Logic-based Benders Decomposition

Master Problem

MIP/CP

Variables:

$$\varphi_{k,r} = \begin{cases} 1, & \text{if request } r \text{ is assigned to vehicle } k \\ 0, & \text{otherwise} \end{cases}$$

$$y_{k,r} = \begin{cases} 1, & \text{if location } i \text{ is visited by vehicle } k \\ 0, & \text{otherwise} \end{cases}$$

Relaxations:

- All sequencing constraints
- All capacity constraints

Subproblem

CP

Variables:

$$\begin{aligned} x_i &= \text{time interval in which location } i \text{ is served} \\ &\quad \text{and is not present if not visited} \\ v_i &= \text{load of the vehicle after visiting location } i \\ u &= \text{sequence of locations visited by the vehicle} \end{aligned}$$

Datasets

75 generated datasets

Number of Nodes	Number of Vehicles	Number of Requests
16	2	6
26	3	10
50	5	20
80	10	30
140	20	50

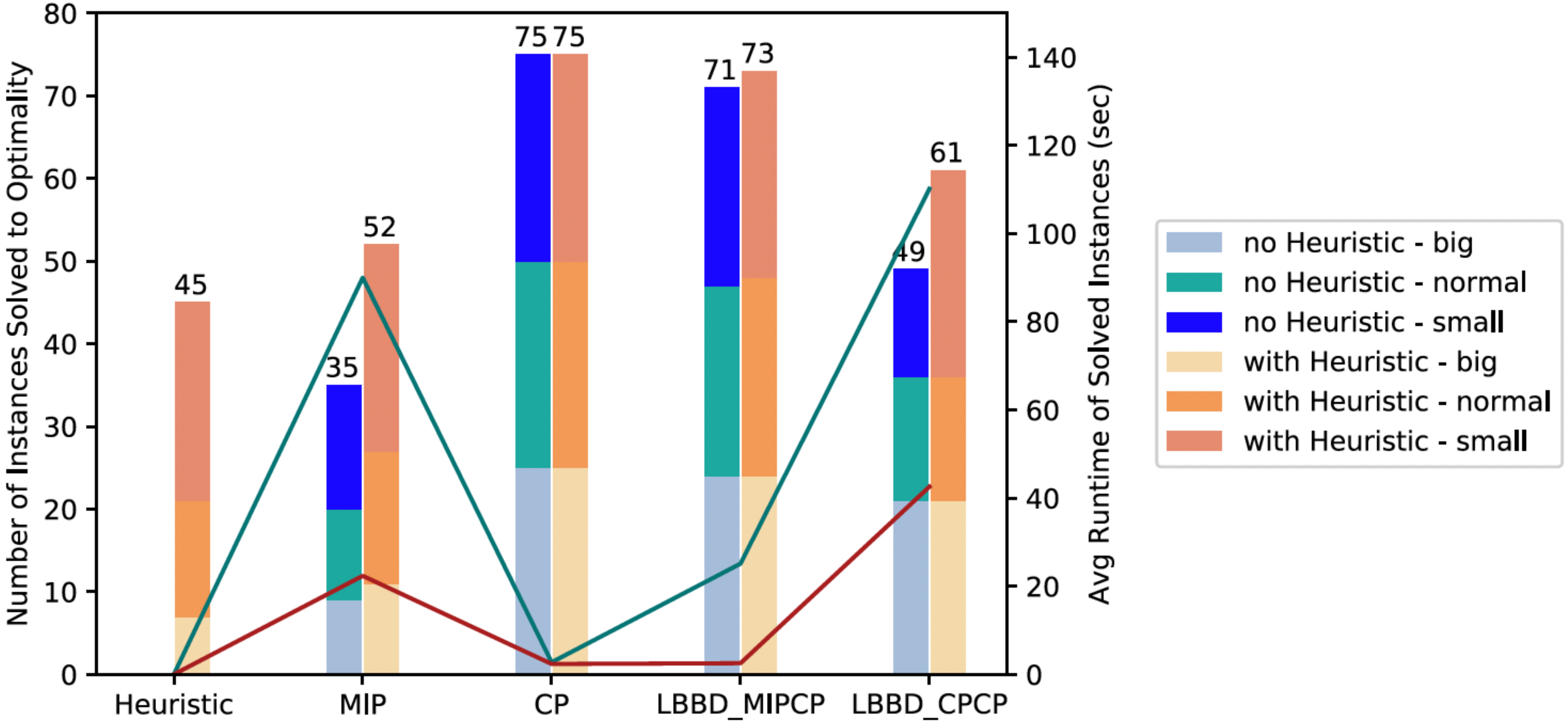
280 real-world instances

	Number of Vehicles	Number of Requests
Average	187	259
Smallest	111	77
Largest	409	591

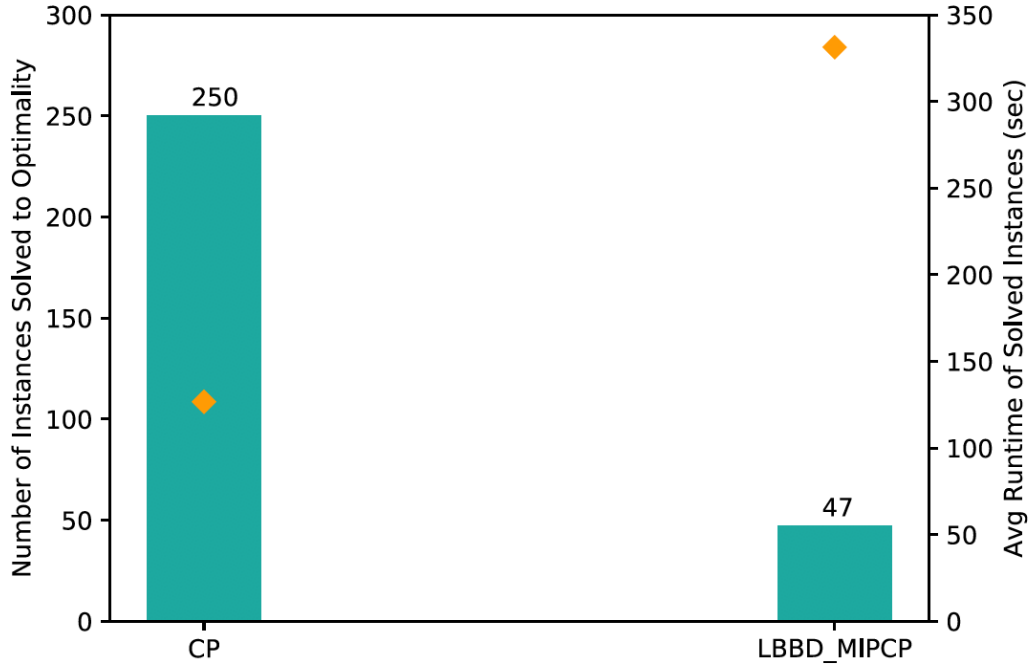
Experimental Setup

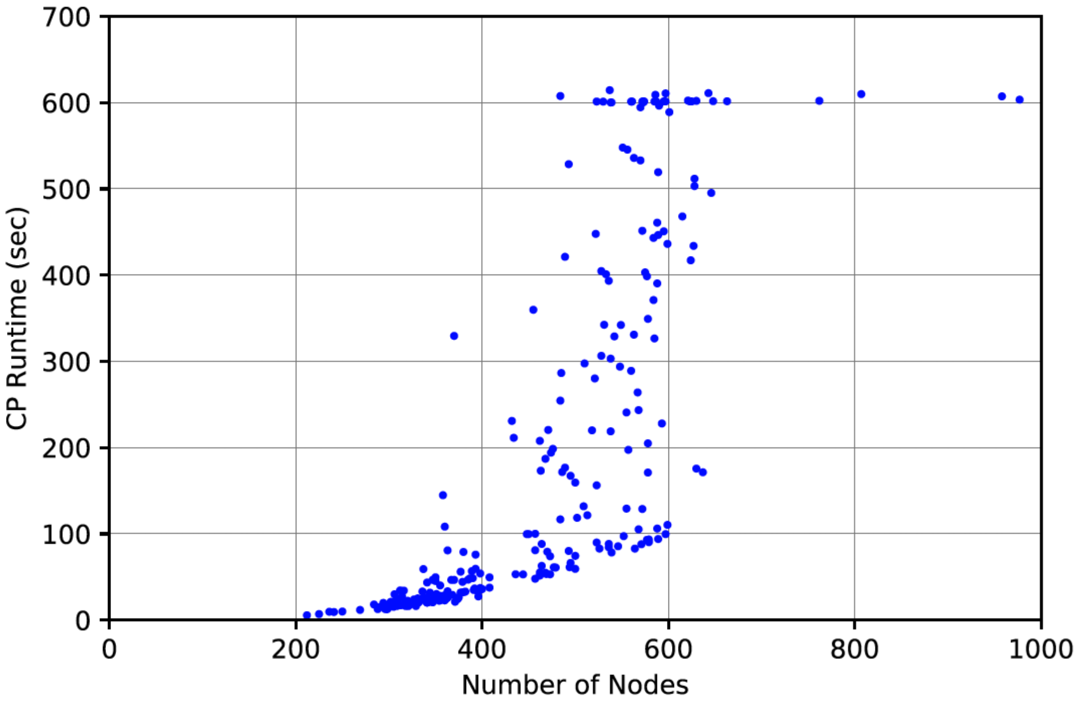
- IBM’s Cplex Studio 12.7 in C++
- Intel Xeon E3-1226 v3 @ 3.30GHz with 16GB RAM
- Single thread
- Time limit: 600 secs

Experimental Results for Generated Datasets



Experimental Results for CHATS Datasets





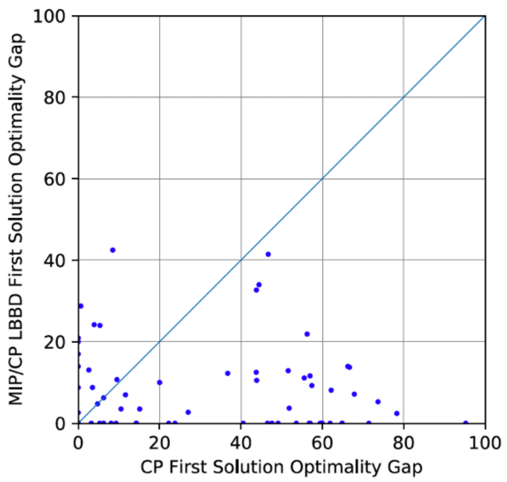
Analysis

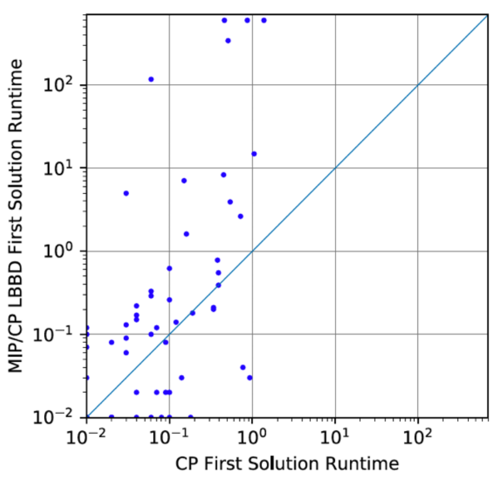
1. The default search of CP Optimizer is particularly suited to our problems

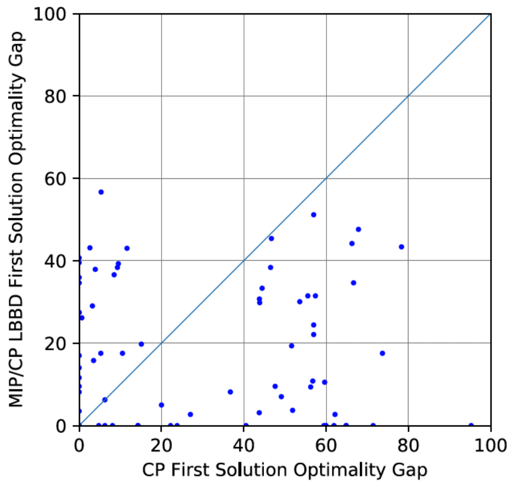
2. CP finds better first solutions

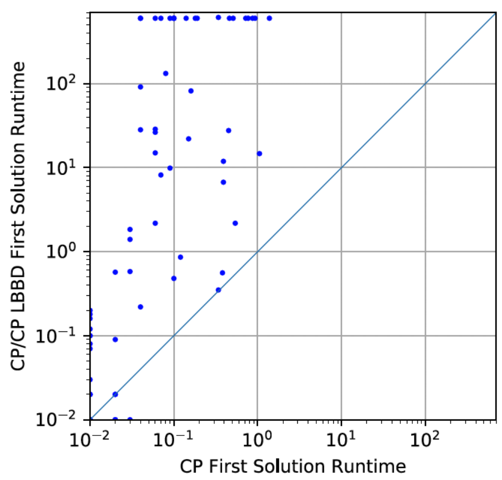
3. Greater impact on search space reduction

First Solution Time



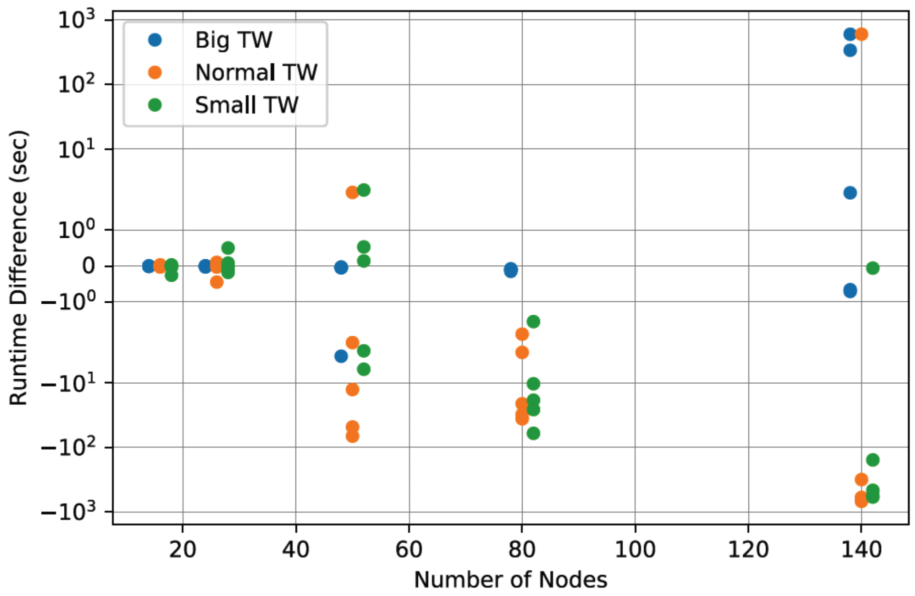




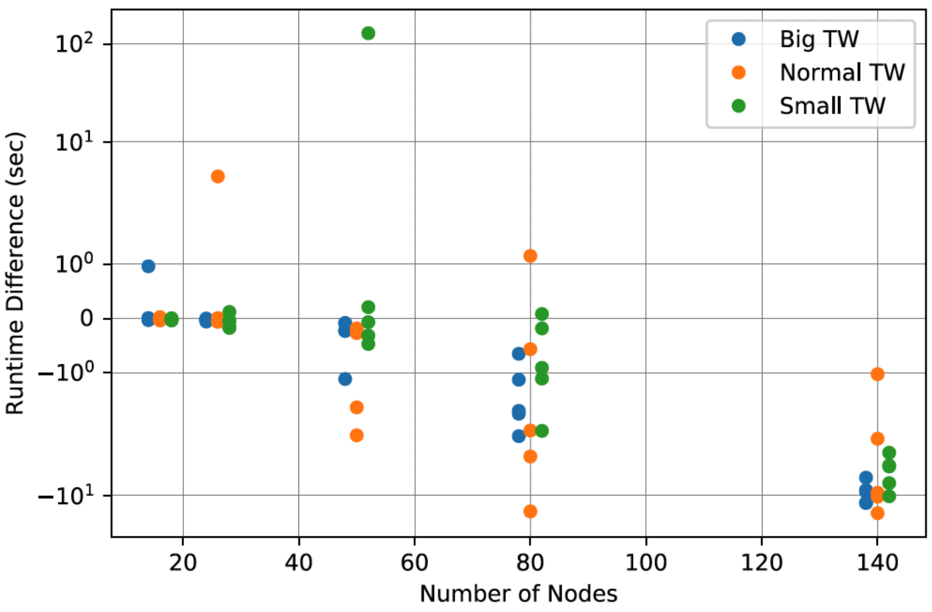


First Solution Quality

Runtime difference of pure MIP/CP LBBD minus MIP/CP LBBD with CP starting solution



Runtime difference of pure CP minus CP with MIP/CP LBBD starting solution



Search Space Reduction

$$\log(|P|) = \log(|D_{x_i}|) + ... + \log(|D_{x_n}|)$$

	TW Type	Lower Bound Percentage				
		100%	80%	60%	40%	20%
CP	small	8 (9.53%)	3 (23.56%)	2 (34.55%)	0 (-)	0 (-)
	normal	3 (1.50%)	1 (0.30%)	0 (-)	0 (-)	0 (-)
	big	0 (-)	0 (-)	0 (-)	0 (-)	0 (-)
CP/CP LBBD	small	1 (5.61%)	1 (5.61%)	0 (-)	0 (-)	0 (-)
	normal	0 (-)	0 (-)	0 (-)	0 (-)	0 (-)
	big	0 (-)	0 (-)	0 (-)	0 (-)	0 (-)

Conclusions

- Proposed a novel problem definition for the Senior Transportation Problem (STP)
- Developed four exact methods to solve the STP and a construction heuristic
- CP shows the most promising performance to solve the STP
  - Good first solution quality and time
  - More impact on the search space reduction

Thank You